

AI-Powered Personalized Sing and Speak for Kids (SS4Kids) for Children with Communication Challenges

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Generative AI is increasingly entering family life, yet relatively little research has examined how AI can support collaborative learning between children and caregivers in ways that strengthen family relationships while promoting child development. This position paper presents ongoing work on AI-powered Personalized Sing and Speak for Kids (SS4Kids), a family-centered speech and language learning platform that enables children and caregivers to co-create personalized songs from meaningful photographs. Designed for young children with communication delays, the system combines evidence-based music therapy principles with constrained generative AI to create individualized communication activities while maintaining caregiver oversight and privacy protections. Rather than positioning AI as an autonomous tutor, we conceptualize AI as a collaborative creative partner that augments caregiver-child interactions and supports repeated communication practice in everyday family contexts. We discuss how this work contributes to emerging conversations on family-centered AI by highlighting design considerations around personalization, shared creativity, caregiver agency, trust, and responsible AI. We hope to engage workshop participants in discussing how AI can better support intergenerational learning, communication, and care within diverse family ecosystems.

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1 Introduction

Families play a central role in children's communication development, particularly for children with communication delays [4]. Caregivers provide the everyday interactions through which children practice language, reinforce therapeutic goals, and build meaningful relationships. As generative AI increasingly enters homes and learning environments, there is an opportunity to design family-centered AI that strengthens caregiver-child collaboration rather than replacing it [3]. Yet relatively little research has explored how AI can support shared communication, creativity, and learning within families while remaining responsible, trustworthy, and developmentally appropriate [7].

Speech and language delays affect millions of young children during the early years of development, when communication skills are most critical for learning, socioemotional well-being, relationships, and lifelong development. Before the COVID-19 pandemic, an estimated 8–10% of children under age eight experienced speech and language delays [2]. The pandemic further

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widened this gap [10], while access to timely, high-quality intervention remains constrained by workforce shortages, long waitlists, and a lack of personalized learning experiences [1]. Children with communication delays, including autistic children, often show strong engagement with multi-modal, interactive digital applications [8]. Yet most speech-language intervention materials rely on prebuilt stimuli (e.g., flashcards and worksheets) with limited opportunities for child-centered personalization or creative ownership, making it difficult for families to sustain children's motivation for repeated communication practice across home, therapeutic, and educational settings.

Music-based speech-language interventions have shown promise for improving communication in young children with speech and language delays [11]. By pairing words and phrases with rhythm and melody, these interventions enhance attention, sustain engagement, and facilitate language learning [12]. Although generative AI (GenAI) and large language models (LLMs) are increasingly being adopted to support learning, their application in family-centered communication interventions remains limited [5, 9]. AI can efficiently create personalized, adaptive learning experiences tailored to each child's interests and communication goals. However, the greatest opportunity lies not simply in personalization, but in designing AI that enables caregivers and children to learn, create, and communicate together.

This gap highlights the need for family-centered AI systems that support shared creativity, communication, and caregiving while maintaining adult oversight, privacy, and developmental appropriateness.

To address this gap, we investigate how generative AI can support family-centered communication learning by enabling children and caregivers to transform meaningful family photographs into personalized songs and language-learning experiences. We build upon Sing and Speak for Kids (SS4Kids), an evidence-based speech-language intervention platform previously supported by the U.S. Department of Education SBIR program, the National Science Foundation (NSF), and the Gates Foundation. SS4Kids is founded on the premise that speech, language, and music share common perceptual mechanisms in young children, enabling therapeutic activities that embed speech stimuli within musical patterns to promote communication development.

Rather than positioning AI as an autonomous tutor, we conceptualize AI as a collaborative creative partner that augments caregiver-child interactions and supports repeated communication practice through meaningful family experiences.

Building on this platform, we integrate GenAI to create personalized songs, language activities, and interactive learning experiences tailored to each child's interests, communication profile, and developmental goals while preserving caregiver agency throughout the process. To ensure these AI-powered features are effective, safe, and practical for real-world family use, the platform is being developed iteratively to examine the following research questions:

RQ1 (Design): What design principles support AI-powered creative learning experiences that are engaging, family-centered, and effective for children with communication delays? **RQ2 (Family Interaction):** How do AI-generated personalized songs and images influence caregiver-child interaction, family participation, communication, and repeated speech practice? **RQ3 (Outcomes):** How does family-centered AI personalization affect children's communication development and family engagement? **RQ4 (Implementation):** What factors influence the responsible adoption of family-centered AI across homes, therapeutic settings, and schools?

This work contributes to the emerging conversation on family-centered AI by generating evidence-based design principles for AI systems that support shared creativity, communication, and caregiving. It advances empirical understanding of how AI-powered learning through play and music shapes family participation while providing practical guidance for researchers, designers, speech-language pathologists, educators, and technology developers designing responsible AI for children and families.

2 AI-powered Personalized SS4Kids: A Family-Centered AI Platform for Communication Through Sing and Play

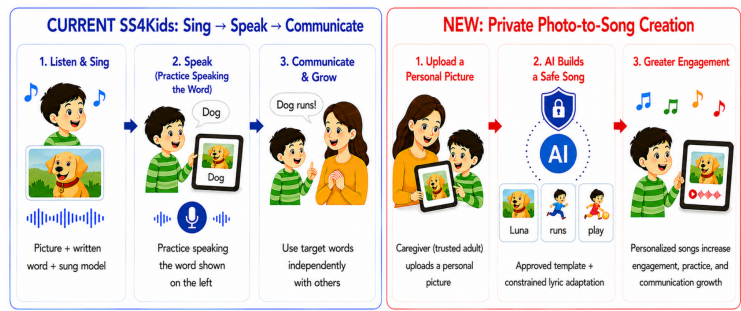


Fig. 1. AI-Powered Personalized SS4Kids: A Family-Centered Communication Platform)

AI-powered Personalized SS4Kids is a family-centered AI platform that enables children and their trusted adults to transform meaningful family photographs into personalized speech and language learning experiences (see Figure 1). Families create a private visual vocabulary library from images representing the child’s everyday life, while constrained generative AI converts selected words into personalized songs that preserve SS4Kids’ evidence-based sing–speak–communicate instructional sequence.

Caregivers remain in control throughout the process. After uploading a photograph, the system suggests a vocabulary label, which the caregiver confirms or edits before selecting the target language and learning goal. AI then generates personalized lyrics and music while maintaining vocabulary targets, repetition, developmental appropriateness, and instructional scaffolding. Children progress through structured activities that move from listening and singing to independently producing and using target words during interactions with caregivers, speech-language pathologists, or teachers. Uploaded images remain private, require adult approval, and are never used to train external AI models.

Rather than functioning as an open-ended AI tutor, Personalized SS4Kids is a protected creative environment where families use their own photographs to create meaningful communication activities together. We view personalization as a family process rather than an individual one. Family photographs, caregiver-selected vocabulary, and shared everyday experiences become resources through which AI facilitates communication, creativity, and relationship building. In this way, AI supports—rather than replaces—the caregiving practices through which families naturally scaffold children’s communication and learning.

Building on prior support from the UC Irvine Proof of Product Grant and NSF VITAL semifinalist recognition for StoryAI [6], the platform also serves as a research testbed for understanding how generative AI can responsibly support creativity, communication, learning through play, and family participation.

3 Discussion Points for the Workshop

As generative AI becomes increasingly embedded in family life, we see AI-powered Personalized SS4Kids as one example of how AI can be designed to strengthen, rather than replace, caregiver-child interactions. We hope to engage workshop participants in discussing broader questions about family-centered AI that extend beyond speech-language intervention.

Specifically, we seek discussion around the following questions:

- Designing family-centered AI. What principles should guide the design of AI systems that support families as collaborative units of learning, communication, and care, rather than focusing solely on individual users?
- Caregiver agency and responsibility. How can AI systems maintain meaningful caregiver oversight while providing adaptive and personalized experiences for children? What roles should caregivers, educators, and clinicians play in AI-mediated learning?
- Personalization through shared family experiences. How can AI responsibly leverage family photographs, memories, routines, and everyday interactions to create meaningful learning experiences while protecting privacy and respecting family values?
- Interdisciplinary perspectives on family-centered AI. How can CSCW, HCI, AI in Education, Child-Computer Interaction, speech-language pathology, and family studies contribute complementary perspectives to designing AI that supports children's development within family ecosystems?
- Future directions for responsible family-centered AI. What research methods, evaluation frameworks, and governance approaches are needed to ensure that AI technologies promote trust, inclusion, accessibility, and well-being for diverse families?

We hope these discussions will help shape a shared research agenda for family-centered AI that recognizes families not simply as end users, but as active collaborators in the design, adoption, and responsible use of AI technologies.

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